Multi-Objective Sparse Subspace Clustering for Hyperspectral Imagery

Yuting Wan, Yanfei Zhong, *Senior Member, IEEE*, Ailong Ma, and Liangpei Zhang, *Fellow, IEEE*

***Abstract*— Hyperspectral images (HSIs) are typical high-dimensional and complex data. As such, the clustering of HSIs is a challenging task. Out of the motivation to find the low-dimensional structure representation of the high-dimensional data, sparse subspace clustering (SSC) methods have been proposed in recent studies. Sparse representation is an important technique in SSC, which is aimed at obtaining the sparse coefficient matrix of the HSI data. Generally speaking, the acquisition of the sparse coefficient matrix is an ill-posed problem, and the existing methods introduce an extra condition as a regularization term to resolve it. However, the regularization parameter is determined manually, which is difficult and lacks self-adaptability. Hence, in this article, a multi-objective SSC method for hyperspectral imagery is proposed, which simultaneously optimizes the sparse term and the data fidelity term. In addition, the spatial structure information of the HSIs is often neglected in the processing model, and thus, a spatial prior term, as the third optimization objective function, is also tested in this article. As a result, there is no need to manually set a regularization parameter. Furthermore, by using the l_0 norm as the sparse term, this reduces the error caused by the convex relaxation of the other norms. In the proposed method, a multi-objective optimization model is first used to acquire the sparse coefficient matrix, in which a strategy for constructing the dictionary is proposed for more precise and efficient multi-objective optimization. In addition, a knee point-based selection method is utilized to automatically select the optimal sparse representation solution from the Pareto front. The adjacency matrix is then constructed according to the sparse coefficient matrix. Finally, a spectral clustering method is used to obtain clustering results. Experiments undertaken with four HSI data sets confirm the effectiveness of the proposed method.**

***Index Terms*— Hyperspectral image (HSI), multi-objective optimization, sparse subspace clustering (SSC).**

I. Introduction

HYPERSPECTRAL remote sensing technology has been an important part of the Earth observation field since the 1980s. It is a multidimensional information acquisition technique that combines imaging and spectroscopy methods [1], [2]. Hyperspectral sensors are used to image a region with hundreds of spectral channels, and thus hyperspectral images (HSIs) have a nearly continuous spectral range [3], [4]. Compared with the traditional multispectral images, HSIs can be used to identify subtle features. As a result, HSIs are widely used in various applications, such as precision agriculture [5], target detection [6], environmental monitoring [7], mineral exploration [8].

HSI classification is the process of assigning land-cover classes to pixels, and the methods used for HSI classification can be categorized as supervised or unsupervised approaches. There are many effective supervised classification approaches, including conditional random field-based methods [9], deep belief network-based methods [10], and so on. However, the supervised approaches need training samples to obtain a high classification accuracy. In addition, the performance of the classifier becomes worse as the number of bands increases (the Hughes phenomenon) [11]. Therefore, clustering methods (i.e., unsupervised classification) have gradually attracted the interest of researchers because they rely only on the inherent structural features of the image for the classification, and they do not require labeled samples [12], [13].

However, in the clustering of high-dimensional data, the feature points are extremely sparse in the feature space, due to the high dimensionality of the feature space. As a result, the traditional method of directly using a distance metric often fails. Based on this analysis, it is natural to explore the low-dimensional structure representation of the high-dimensional data. In fact, in many practical problems, such as handwritten character recognition [14], moving objects in videos [15], face image clustering [16], and so on, the data can be represented in a low-dimensional subspace of the enclosed space. What is more, the multi-class data often lie in a union of the low-dimensional subspaces [17]. Thus, a series of high-dimensional data clustering methods based on sparse subspace clustering (SSC) have been proposed. In 2009, Elhamifar and Vidal [18] proposed an SSC method based on 1-D sparsity. In 2010, Liu *et al.* [19] further proposed a subspace clustering method based on low-rank representation (LRR) using 2-D sparsity. Subsequently, the research has mainly focused on the design of the regularization term and data fidelity term of the algorithms [19]–[22]. The regularization term is aimed at ensuring that the sparse coefficient matrix has a good structure that is favorable for subspace clustering, and the data fidelity term is used to make the algorithm robust to various types of noise.

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The authors are with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan 430079, China, with the Hubei Provincial Engineering Research Center of Natural Resources Remote Sensing Monitoring, Wuhan University, Wuhan 430079, China, and also with the Collaborative Innovation Center of Geospatial Technology, Wuhan University, Wuhan 430079, China (e-mail: wanyuting@whu.edu.cn; zhongyanfei@whu.edu.cn; maailong007@whu.edu.cn; zlp62@whu.edu.cn).

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Recently, SSC methods have been applied for hyperspectral anomaly detection [23] and endmember extraction [24], and they have also been introduced into HSI clustering. The SSC methods for HSIs assume that the pixels in each class lie in a subspace. In 2016, SSC was first utilized in HSI clustering by Zhang *et al.* [25], who proposed spectral-spatial SSC. Subsequent developments in HSI clustering include the l_2-norm regularized SSC algorithm [26], the kernel SSC method with a spatial max pooling operation [27], and the spatial distribution preserving-based SSC method [28]. In addition, Yan *et al.* [29] designed a discriminated similarity matrix based on SSC for HSI clustering. Generally speaking, SSC can be regarded as an ill-posed problem, due to the fact that we want to make the sparse matrix as sparse as possible while minimizing the amount of missing data. In order to address this problem, the common solution is to introduce additional information as a prior to regularize the ill-posed problem, where the additional information can be used to regularize the relationship between the sparse term and the data fidelity term. However, the determination of the regularization parameter is usually set beforehand by a trial-and-error strategy. Furthermore, the regularization parameter is dependent on the remote sensing data and cannot be determined adaptively [30]. From another perspective, when the regularization parameter does not need to be determined manually, this tends to result in multiple trade-off solutions for the sparse representation coefficient matrix. Thus, the process of sparse optimization can be modeled as a multi-objective optimization problem. This approach eliminates the process of manual subjective determination of the regularization parameter, and the task is completed by machine instead of the people. However, due to the limitations of the external conditions and the imaging resolution, the spatial structure of the image also needs to be taken into consideration. In this article, in order to consider the spatial correlation of the pixels, the total variation (TV) prior term [65] is designed as a spatial prior term and is regarded as the third optimization objective function. Multi-objective evolutionary algorithms (MOEAs) have the ability to optimize multiple objective functions simultaneously and generate a set of trade-off solutions [31]. In addition, EAs were first proposed in the 1980s, and are random search algorithms based on population-based simulation of natural evolution. An EA searches for the solution or solution set of the black box problem by an iterative loop, without any prior knowledge. The general paradigms of an EA are as follows: the population is randomly initialized, and the offspring population is generated through the parent population in the process of each iteration. The best individual is then selected as the next generation of the parent population through the fitness function until the termination condition is reached [32]. EAs having the capability of global search, have gradually become an effective tool for solving multi-objective optimization problems [33]–[37]. After nearly 30 years of development, the existing MOEAs can be roughly divided into three categories [38]: 1) Pareto-based methods; 2) index-based methods; and 3) decomposition-based methods. Hence, in this article, in order to avoid the selection of the regularization parameter, an MOEA is utilized to balance the sparse term and the data fidelity term.

MOEAs have been widely utilized in remote sensing image analysis and interpretation [39], [40], in applications, such as image clustering [41]–[43], HSI band selection [44], sparse unmixing of HSIs [45], [46], change detection [47], [48], subpixel mapping [49], and so on. For sparse unmixing and subpixel mapping, in particular, MOEAs are utilized to simultaneously optimize the data fidelity term, the sparsity term, and the TV regularization term. From these successful applications, it can be seen that MOEAs have the potential to deal with the above ill-posed problems, which motivates us to utilize an MOEA for the ill-posed SSC problem for hyperspectral imagery.

In this article, a multi-objective SSC framework for hyperspectral imagery is proposed, in which the MOEA is utilized to obtain a better sparse coefficient matrix through simultaneously optimizing the sparse term and the data fidelity term. In addition, as a model expansion, the TV spatial prior term is introduced to conduct three-objective SSC of hyperspectral imagery. In order to achieve this, a dictionary construction strategy is designed, and a knee point-based selection strategy is utilized. The main contributions of this article are described as follows.

1) *A Multi-Objective SSC Framework for Hyperspectral Imagery With More Accurate Modeling:* A multi-objective SSC framework is proposed for hyperspectral imagery. In the proposed method, the nondominated sorting genetic algorithm II (NSGA-II) [50] is used for bi-objective optimization (the sparse term and the data fidelity term), and three-objective optimization (with the introduction of the TV spatial prior term) is used in the SSC model for hyperspectral imagery. As a result, there is no need to manually determine a regularization parameter in advance, which makes the method more adaptive. In order to achieve more accurate modeling of the multi-objective sparse optimization, and considering that MOEAs can be used to solve nonconvex optimization problems, we directly optimize the l_0 norm as the sparse matrix for the SSC of hyperspectral imagery, which allows more accurate modeling and reduces the error caused by the convex relaxation of other norms, such as the l_1 norm. In addition, an experimental method based on the optimization of the l_1 norm was also designed for a specific comparison. Furthermore, the TV spatial prior term is also modeled in the proposed method, to consider the spatial structure of the HSIs.
2) *The Designed Strategies for the Effective SSC of Hyperspectral Imagery:* How to achieve effective and automatic processing is one of the most important aspects of HSI clustering. In the proposed method, a dictionary construction strategy and knee point-based selection strategy are designed and introduced for multi-objective SSC. The accuracy and efficiency of multi-objective optimization are often sensitive to the length of the individuals. However, in the existing SSC methods for hyperspectral imagery, the dictionary is the

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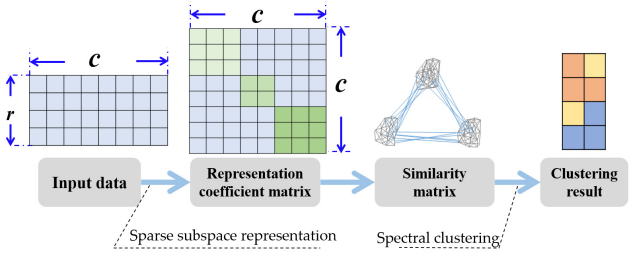


Fig. 1. General flowchart of the SSC algorithm.

2-D expression of the original HSI data, which results in the acquired dictionary being huge, redundant, and inefficient. In the proposed method, in order to resolve this problem, a dictionary construction strategy is achieved by preclustering and then selecting pixels with a maximum membership degree. A Pareto optimal solution set for the sparse optimization matrix can be acquired after the multi-objective sparse optimization. However, only one sparse optimization matrix is needed as the final optimal result. In the proposed method, the knee point-based selection strategy is used to automatically select the final optimal sparse optimization result from the final optimal solution set, which has not previously been achieved for the multi-objective SSC of HSIs.

The rest of this article is organized as follows. Section II gives the related background, including SSC and multi-objective optimization. Section III describes the framework of the proposed approach. The experiments and analyses are presented in Section IV. The parameter sensitivity analyses are provided in Section V. Finally, the conclusion is drawn in Section VI.

II. Background

A. SSC

In the field of subspace clustering, there are five main types: 1) matrix decomposition-based methods, 2) algebraic methods, 3) statistical methods, 4) iterative methods, and 5) spectral clustering methods [51]. The SSCs are essentially spectral clustering methods, have recently become a research hotspot.

1) Sparse Subspace Representation: As shown in Fig. 1, the basic idea of SSC is to represent the point as a linear combination of the other data points, which can be expressed as

$$y_i = \mathbf{D}^{r\times c} \times c_i \tag{1}$$

where y_i represents the data to be represented, $\mathbf{D}^{r\times c}$ is the data dictionary, and c_i is the coefficient vector. It should be noted that the 2-D data matrix $\mathbf{D}^{r\times c}$ is expressed as the dictionary for the sparse subspace representation. In other words, this data matrix is a self-expressive dictionary. r is the dimension, and c represents the total number of feature points.

We convert (1) into a matrix form, as shown in the following:

$$\mathbf{Y} = \mathbf{DC} + \mathbf{E} \quad \text{s.t. } \text{diag}(\mathbf{C}) = 0 \tag{2}$$

where $\mathbf{Y}$ represents the original data matrix, $\mathbf{C} = \{c_1, c_2, \ldots, c_n\}$ is the sparse coefficient matrix, $\mathbf{E}$ denotes the representation error matrix, and $\mathbf{C}_{ii} = 0$ is utilized to avoid a trivial solution. Normally, the data dictionary is an overcomplete dictionary, which leads to the existence of many feasible solutions. By referring to the sparse criterion [52], the sparse coefficient matrix can be obtained by optimizing

$$\min \|\mathbf{C}\|_0 \quad \text{s.t. } \mathbf{Y} = \mathbf{DC} + \mathbf{E}, \ \text{diag}(\mathbf{C}) = 0 \tag{3}$$

where the l_0 is the number of nonzero elements.

However, the l_0-norm is a nonconvex problem, so the l_1-norm is often used to approximate convex optimization

$$\min \|\mathbf{C}\|_1 \quad \text{s.t. } \mathbf{Y} = \mathbf{DC} + \mathbf{E}, \ \text{diag}(\mathbf{C}) = 0 \tag{4}$$

where the l_1-norm represents the sum of the absolute values of the elements in a vector or matrix.

Actually, in many problems, data in a class can be well represented by a low-dimensional subspace of the high-dimensional ambient space, due to the data within a class having a better aggregation effect than the data in different classes [18]. Furthermore, the successful applications, such as handwritten character recognition, face image clustering, and so on, have confirmed the effectiveness of the SSC method. Thus, for HSIs, as typical high-dimensional data, we want to try and prove the validity of low-dimensional representation by using an SSC method. In addition, the "subspace" refers to a subset of the full space, and an ideal subspace wants to make the existing sample points belong to the same class as much as possible. For HSI clustering, the pixels of the same land-cover type need to be included in the same subspace. Thus, it can be found that the high-dimensional space S_{full} can be a combination of multiple linear subspaces S_i, as shown in the following:

$$S_{\text{full}} = \bigcup_{i=1}^{n} S_i. \tag{5}$$

2) Constructing the Adjacency Matrix and Spectral Clustering: It should be noted that the sparse optimization procedure is used to give the subspace-based sparse representation for each sample point; in other words, a nonzero element corresponds to a representation of a point from the same subspace of the given data points. However, assuming that we have two data points $\mathbf{y}_m$ and $\mathbf{y}_n$, some sample points, including $\mathbf{y}_n$, can be written as a linear combination for $\mathbf{y}_m$, while the sparse representation of $\mathbf{y}_n$ may not choose $\mathbf{y}_m$. Through this special selection of weight, we ensure that nodes m and n are interconnected, and the similarity is then calculated. The adjacency matrix $\mathbf{W}$ is then constructed as

$$\mathbf{W} = \mathbf{C} + \mathbf{C}^T. \tag{6}$$

The final clustering results are then obtained by applying spectral clustering to the Laplacian matrix induced by the adjacency matrix $\mathbf{W}$.

B. Formulation of the Problem and Motivation

Sparse optimization in SSC involves modeling the relationship between the sparse matrix term and the representation error term, which is an ill-posed problem. In order to resolve this problem, the traditional methods take an additional condition into consideration, and a regularization parameter is

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introduced, as shown in the following:

$$\min_{\mathbf{C}}(1/2)\|\mathbf{Y}-\mathbf{DC}\|_F^2+\rho\|\mathbf{C}\|_1 \quad \text{s.t. } \operatorname{diag}(\mathbf{C})=0 \tag{7}$$

Since an optimization problem can be converted into T independent subproblems (8), the alternating direction method of multipliers (ADMM) [53] can be used to solve a convex optimization problem by decomposing the convex optimization problem into smaller parts, which correspond to the subproblems and it can make the problem easier to handle. Thus, we can apply the ADMM (9) to solve this problem

$$\min \ (1/2)\|y_i-\mathbf{D}c_i\|_2^2+\rho\|c_i\|_1, \quad i=1,2,\ldots,T \tag{8}$$

$$\begin{aligned}&\arg\min_{\mathbf{C},\mathbf{Z}}\left\{\frac{1}{2}\|\mathbf{Y}-\mathbf{DC}\|_F^2+\rho\|\mathbf{Z}\|_1+\frac{Q}{2}\|\mathbf{C}-\mathbf{Z}\|_F^2\right\}\\ &\quad\Rightarrow L(\mathbf{C},\mathbf{Z},\Im)=\frac{1}{2}\|\mathbf{Y}-\mathbf{DC}\|_F^2+\rho\|\mathbf{Z}\|_1\\ &\quad+\frac{Q}{2}\|\mathbf{C}-\mathbf{Z}\|_F^2+\operatorname{vec}(\Im)^T\operatorname{vec}(\mathbf{C}-\mathbf{Z})\\ &\text{s.t. } \mathbf{C}=\mathbf{Z}, \quad \operatorname{diag}(\mathbf{C})=0\end{aligned} \tag{9}$$

where the first term in (9) is the data fidelity term, i.e., the l_2-norm, the second term is the sparse term, and the third term is the Lagrange multiplier term. ρ is the regularization parameter, Q is the Lagrange multiplier parameter, and vec($\Im$) is the vectorized Lagrangian multiplier matrix.

However, this raises the question of how to tune parameter ρ to keep the balance between the data fidelity term and the sparse term, due to it having a great effect on the final results [26]. For the existing problems, we want to reduce or even eliminate the effect of this parameter. Furthermore, the representation error caused by the convex relaxation of the l_0 norm is also nonnegligible.

Thus, it is natural to consider optimizing the data fidelity term $\|\mathbf{Y}-\mathbf{DC}\|_2^2$ and sparse term $\|\mathbf{C}\|_0$ simultaneously, instead of considering the regularization parameter and the l_1 norm $\|\mathbf{C}\|_1$. Hence, at this point, it is a weight parameter-free model, and more accurate modeling is achieved. However, as the data fidelity term and sparse term need to be optimized simultaneously, this is completely different from optimizing a single objective function. Thus, a novel multi-objective optimization strategy is designed as part of the proposed framework. Some research background to the multi-objective optimization approach is presented in Section II-C.

C. Multi-Objective Optimization Approach

Since the data fidelity term, the sparse term, and the TV spatial prior term are regarded as the objective functions for the SSC of hyperspectral imagery, and they need to be optimized simultaneously. For multi-objective optimization, we need to compare the multiple and conflicting objective functions, and we want to keep as much information as possible while making the HSI data as sparse as possible. The notion of Pareto domination is utilized in Fig. 2, to allow a better understanding,. In addition, some basic definitions are provided in the following.

Definition 1 (Multi-Objective Optimization Problem): A general description is presented. Given the decision variables

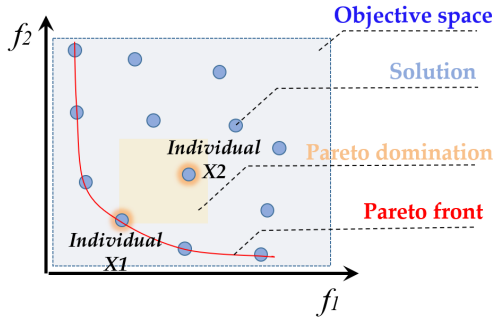


Fig. 2. Pareto domination (X_1 Pareto dominates X_2) and the Pareto front.

$X=(x_1,x_2,\ldots,x_n)$, they satisfy the following constraints:

$$g_i(X)\geq 0, \quad (i=1,2,\ldots,k) \tag{10}$$

$$h_i(X)=0, \quad (i=1,2,\ldots,l). \tag{11}$$

We assume that there are r optimization objective functions, and they are conflicting with each other, as shown in the following equation:

$$\min f(X)=(f_1(X),f_2(X),\ldots,f_r(X)). \tag{12}$$

Thus, we seek the final optimal solutions $X*=(x_1^*,x_2^*,\ldots,x_n^*)$, while satisfying the constraints in (10) and (11).

Definition 2 (Pareto Domination): For the decision variables X_1, X_2, when they satisfy the condition in (13), it can be said that X_1 Pareto dominates X_2, as shown in Fig. 2

$$\begin{aligned}&\forall i=1,2,\ldots,r,\ f_i(X_1)\leq f_i(X_2)\wedge\\ &\exists j=1,2,\ldots,r,\ f_j(X_1)<f_j(X_2).\end{aligned} \tag{13}$$

Definition 3 (Pareto Front): The final result of the multi-objective optimization problem is not a single solution, but a Pareto optimal solution set. As shown in Fig. 2, the final searched solutions form the Pareto optimal solution set, in which the solutions are nondominated with regard to each other. Furthermore, in the multi-objective SSC of hyperspectral imagery, a series of trade-off solutions between the objective functions of the data fidelity term and the sparse term is obtained, which is closer to the actual situation.

Hence, the multi-objective optimization methods aim at obtaining a Pareto front with high-quality solutions. The strategies of nondominated sorting and density estimation ranking used in NSGA-II [50] are utilized in the proposed method. In addition, genetic evolutionary operations are also used to generate more accurate solutions.

III. Multi-Objective SSC for Hyperspectral Imagery

In the proposed algorithm, the use of multi-objective optimization involves performing sparse optimization to obtain a better sparse coefficient matrix, thereby avoiding the need for a regularization parameter and improving the adaptability. For the subsequent construction of the adjacency matrix, a "full connection" method [55] is used, and the Gaussian kernel function [54] from the field of radial basis function (RBF) [55] theory is used for the calculation. Finally, the normalized

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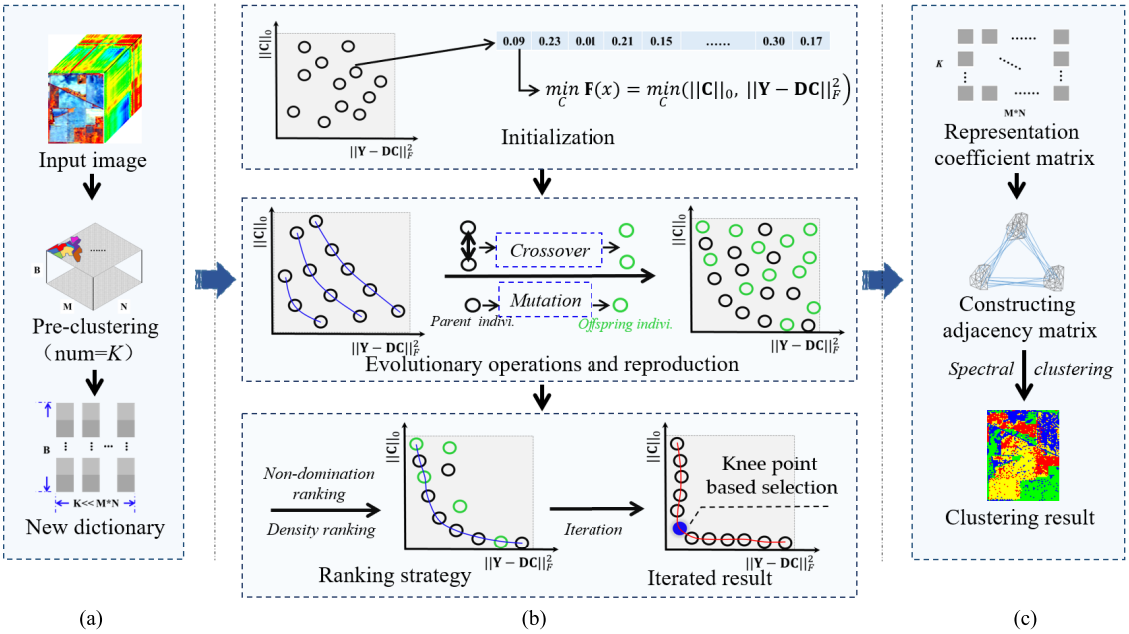


Fig. 3. Basic framework of the proposed method. (a) Dictionary construction and preprocessing. (b) Multiobjective sparse optimization. (c) Spectral clustering and result output.

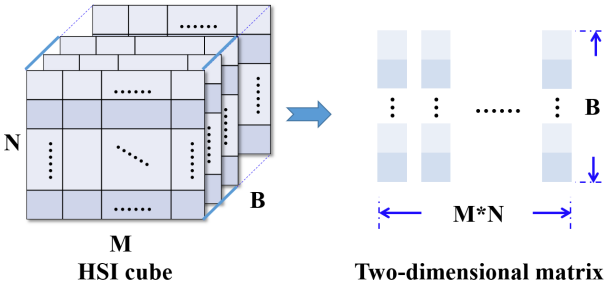


Fig. 4. Traditional method of dictionary construction.

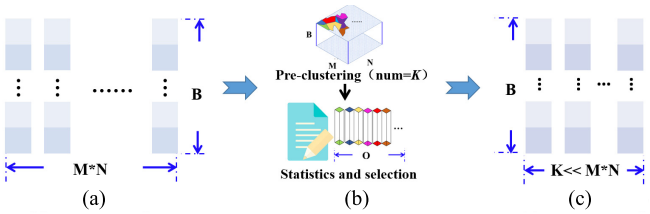


Fig. 5. Dictionary preprocessing. (a) Old dictionary. (b) Dictionary preprocessing. (c) New dictionary.

cut (Ncut) [56] method is used for the spectral clustering, and the final clustering result is returned. The basic framework of the proposed method is shown in Fig. 3.

A. Dictionary Construction and Preprocessing

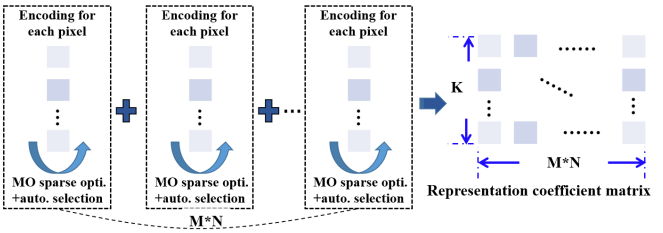
Before the sparse optimization, a data dictionary is required. For an HSI, it is a structure that represents the data in the form of a 3-D cube. Thus, this should be converted into the form of a 2-D matrix (see Fig. 4), where M, N, and B are the number of columns, rows, and bands, respectively.

In the original SSC method, as shown in Fig. 4, during the construction of the dictionary, it can be clearly seen that the number of dictionary columns is built directly from the original data, and is the same as the total number of pixels in the HSI. However, we know that the total number of pixels M × N in the HSI is huge, and the size of the sparse matrix we want is [M × N, M × N]. Thus, the computational complexity of the subsequent sparse optimization is too high. In the experimental analysis part of this article, an execution time comparison is provided to confirm this conclusion. Furthermore, due to the fact that an EA is applied in our multiobjective SSC framework, each sparse representation vector of each pixel under the dictionary is encoded in the individual. However, the length of the individual is equal to M × N, which is a problem of hyper-scale optimization. Hence, this has a great effect on the final sparse optimization results. With this motivation, we want to reduce the length of the individuals, so reducing the number of columns of the dictionary is considered in this article. The specific pretreatment operations are shown in Fig. 5 and are described below.

As shown in Fig. 5(b), fuzzy c-means (FCM) multi-category clustering is first performed on the HSI data, where the number of categories is K. The membership value μ_{ij} of each data point in each cluster is then counted by (14) and (15), and the data point with the highest degree of membership in each cluster is selected. The selected data points are then used to form a new data dictionary matrix. From Fig. 5(c), it can be seen that the number of columns K of the new dictionary is much smaller than M × N. On this basis, the subsequent sparse optimization and evolutionary multi-objective sparse optimization can be conducted

$$\mu_{ij} = \frac{1}{\sum_{k=1}^{K}\left(\frac{1/\|x_j - U_i\|}{1/\|x_j - U_k\|}\right)^{2/(m-1)}}$$
$$\text{s.t.} \sum_{i=1}^{K} \mu_{ij} = 1, \quad j = 1, \ldots, Nu \tag{14}$$

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Note: *MO sparse opti.: multi-objective sparse optimization; auto. selection: automatic selection.*

Fig. 6. Obtaining the sparse coefficient matrix by multiobjective sparse optimization of each pixel.

$$U_i = \frac{\sum_{j=1}^{Nu} u_{ij}^m x_j}{\sum_{j=1}^{Nu} u_{ij}^m} \tag{15}$$

where K is the number of clusters; Nu is the number of whole pixels in the image; x_j is a vector representing the jth pixel of the remote sensing image; μ_{ij} represents the fuzzy membership of the jth pixel, with respect to the ith cluster center; m is a fuzzy weighting exponent on each fuzzy membership μ_{ij}; U_i is the ith cluster center; and U_k is the kth cluster center.

B. Multi-Objective Sparse Optimization

After the new dictionary is obtained, multi-objective sparse optimization is used to obtain the sparse coefficient matrix. The main idea is to use an evolutionary multi-objective algorithm (NSGA-II is used in this article) to optimize the sparse representation of each pixel in the image under the new dictionary, as shown in Fig. 6. However, the result of each iteration of the multi-objective optimization is the Pareto optimal solution set, and one solution needs to be selected as the final solution. In the proposed method, a knee point-based selection strategy [57], [58] is utilized in the multi-objective SSC for hyperspectral imagery. A sparse coefficient matrix can then be formed after the sparse representation of all the pixels. Based on the analyses of Section II-B, the data fidelity term and sparse term are presented, as shown in (16). The purpose of the sparse term $\|\mathbf{C}\|_0$ is to represent the original data with as few elements as possible, to reduce the redundancy, and the data fidelity term $\|\mathbf{Y} - \mathbf{DC}\|_F^2$ is used to minimize the error of the sparse representation. The sparse matrix $\mathbf{C}$ is what we want to acquire. Hence, it can be seen that the data fidelity term and sparse term are contradictory, and both need to be minimized.

$$\begin{aligned} &\min_C \mathbf{F}(x) = \min_C \left(\|\mathbf{C}\|_0, \|\mathbf{Y} - \mathbf{DC}\|_F^2\right) \\ &\text{s.t. } \mathbf{C}_{ij} \geq 0, \quad 1^T\mathbf{C} = 1^T \end{aligned} \tag{16}$$

where the nonnegative $\mathbf{C}_{ij} \geq 0$ constraints are first added. In addition, the constraint $1^T\mathbf{C} = 1^T$ means that an affine subspace model is adopted, which is a special linear subspace model.

In addition, considering the spatial prior, the third optimization objective function is constructed by using the TV spatial prior to realizing the three-objective SSC of hyperspectral remote sensing imagery. The expression is as shown in the following:

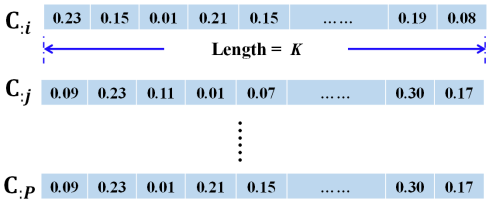


Fig. 7. Individual encoding strategy and population initialization.

$$\begin{aligned} &\min_C \mathbf{F}(x) = \min_C\{f_1(\mathbf{C}), f_2(\mathbf{C}), f_3(\mathbf{C})\} \\ &\quad \text{Sparse term}: f_1(\mathbf{C}) = \|\mathbf{C}\|_0 \\ &\quad \text{Data fidelity term}: f_2(\mathbf{C}) = \|\mathbf{Y} - \mathbf{DC}\|_F^2 \\ &\quad \text{TV prior term}: f_3(\mathbf{C}) = \sum_{i=1}^{M\times N} \|\mathbf{C}_{:i+1} - \mathbf{C}_{:i}\|_1 \\ &\text{s.t. } \mathbf{C}_{ij} \geq 0, \quad 1^T\mathbf{C} = 1^T \end{aligned} \tag{17}$$

where $\|\cdot\|_1$ denotes the l_1-norm.

The evolutionary multi-objective algorithm is essentially a population-based iterative search method [59]–[61]. Thus, a population with a certain number of individuals is required, where the individual represents the sparse representation vector, and the Pareto front denotes a set of nondominated individuals. The evolutionary operations can then be conducted.

1) Initialization: For the initialization of the individual, each sparse vector of the sparse matrix for each pixel in the HSI is encoded as the individual. In addition, the length of the individual depends on the number of categories for preclustering, and length $= K$. The encoding strategy is presented in Fig. 7. Each element in the individual is randomly initialized as $\mathbf{C}_{:j} = \text{rand}(0, 1)$, and the individual also needs to meet the constraints in (16), which means that each column $\mathbf{C}_{:j}$ of the coefficients $\mathbf{C}$ is randomly initialized according to a zero-mean unit-variance Gaussian distribution and then projected back to the unit simplex. In addition, the P individuals in a population are initialized simultaneously.

The values of the objective functions are calculated for each individual. To refine the population, the evolutionary operations are then conducted.

2) Evolutionary Operations and Reproduction: The crossover and mutation operations in the genetic algorithm are utilized, which are analogous to the crossover and mutation of the chromosomes in biology. The specific operations involve randomly selecting two individuals from the population for the multi-point crossover operation and then swapping the corresponding positions of the two individuals. For the mutation, an individual directly mutates at one or more gene positions. The process is shown in Fig. 8.

The values of the two objective functions are then recalculated for each individual. In addition, the size of the population is now $2 \times P$. Thus, a ranking strategy is adopted.

3) Ranking Strategy: Since the number of individuals in a population should be maintained as a constant P, the ranking strategy from NSGA-II is applied. The details are shown in Fig. 9 and presented below.

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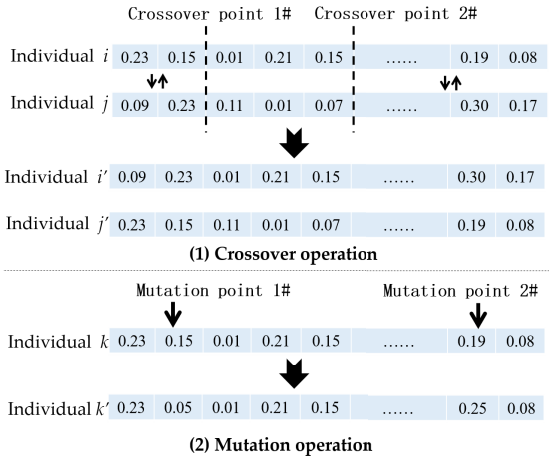


Fig. 8. Crossover and mutation.

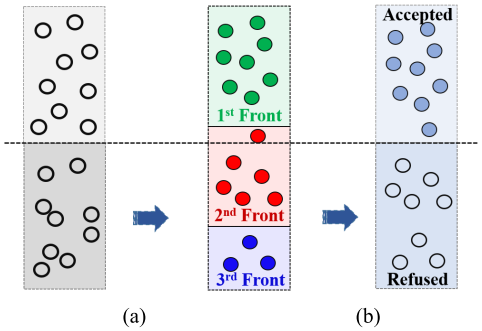


Fig. 9. Ranking strategy. (a) Nondominated soring. (b) Density estimation ranking.

As shown in Fig. 9, nondominated sorting is first used, which uses the idea of Pareto sorting to number the individuals in the population. Individuals in the population that are not dominated by other individuals are first numbered as 1. We can then obtain the first front. After the first front is accomplished, the other individuals are also compared by the idea of Pareto domination, and the second front is obtained. In the same way, we can obtain the third front. The operation is repeated until all the individuals in a population are marked with the Pareto front number. It can be considered that individuals with the same number are nondominated with regard to each other. After the nondominated sorting, it is natural to think that the individuals with a smaller Pareto front number should be considered first. However, under normal circumstances, the number of individuals with the number 1 is often less than or more than P, and thus, the individuals selected as the next generation need to be supplemented or picked. On this basis, a density sorting strategy is utilized, which evaluates the compactness of the individuals. The formula for the crowding distance is expressed as shown in (18), where the individuals holding a larger value are selected to enter the next iteration

$$\begin{aligned} \text{CDist}_i &= \frac{f_1(\mathbf{C}_{:i-1}) - f_1(\mathbf{C}_{:i+1})}{f_1^{\max} - f_1^{\min}} + \frac{f_2(\mathbf{C}_{:i+1}) - f_2(\mathbf{C}_{:i-1})}{f_2^{\max} - f_2^{\min}} \\ \text{CDist}_i^{\text{3-TV}} &= f_1\{\ldots\} + f_2\{\ldots\} + \frac{f_3(\mathbf{C}_{:i+1}) - f_3(\mathbf{C}_{:i-1})}{f_3^{\max} - f_3^{\min}} \end{aligned} \tag{18}$$

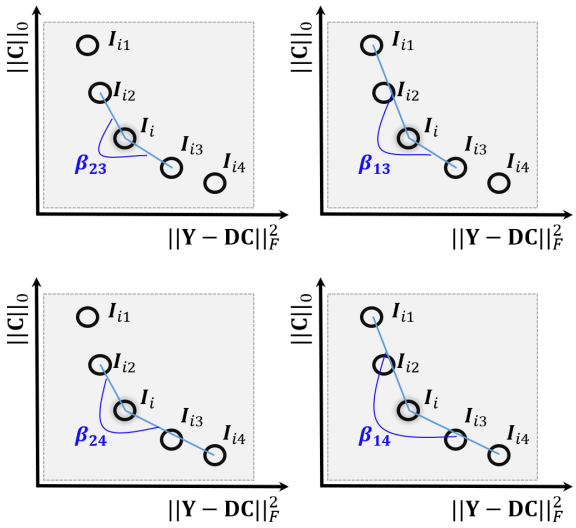


Fig. 10. Knee point-based selection for the final solution.

where CDist_i represents the bi-objective optimization for the SSC of hyperspectral imagery; $\text{CDist}_i^{\text{3-TV}}$ represents the three-objective optimization; $f_1(\cdot)$, $f_2(\cdot)$, and $f_3(\cdot)$ represent the objective functions, which are denoted as shown in (16) and (17). $f_1^{\max}$, $f_1^{\min}$, $f_2^{\max}$, $f_2^{\min}$, $f_3^{\max}$, and $f_3^{\min}$ are the maximum and minimum values of the three objective functions in a population.

4) Automatic Selection of the Solution: This is undertaken after a certain number of generations for evolutionary operations and ranking. The solutions obtained by the multi-objective optimization form a Pareto optimal solution set, and they are nondominated with regard to each other. Thus, a final solution should be selected as the final sparse vector for the sparse representation of a pixel in the HSI. The knee point-based method is used in this article, as shown in Fig. 10.

Generally speaking, the knee point of the optimal solution set of a multi-objective optimization problem is considered to be the position of the unique optimal solution. In the optimal solution set, the knee point has the following characteristic: a slight change of one objective function will cause a dramatic change in another objective function [62], [63]. As shown in Fig. 10, in order to select the knee point, four neighbors are considered. For individual I_i, the four neighbors are considered to calculate the four angles (β_{23}, β_{13}, β_{24}, and β_{14}). Thus, every individual has the maximum reflex angle computed from its neighbors, which is considered as the trade-off solution. The global optimal trade-off solution can then be obtained by comparing all the maximum angles for all the individuals in the Pareto optimal solution set.

5) Stopping Condition: The above multi-objective sparse optimization is performed on each pixel of the HSI to obtain the sparse representation vector, as shown in Fig. 3(B). The sparse representation matrix $\mathbf{C}_{ij}$ is then constructed, where $i \in \{1, \ldots, K\}$, $j \in \{1, \ldots, M * N\}$.

C. Spectral Clustering and Result Output

The sparse representation matrix $\mathbf{C}_{ij}$ is obtained through the evolutionary multi-objective sparse optimization, and the

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adjacency matrix can then be obtained by the sparse representation matrix. However, $\mathbf{C}_{ij}$ is not a square matrix, and thus, the adjacency matrix is different from the former $\mathbf{W} = \mathbf{C}+\mathbf{C}^T$. A commonly used operation in spectral clustering known as the full connection method [55] is used in the proposed method, and the Gaussian RBF kernel [54] is used for the calculation of the adjacency matrix. As shown in (19) and Fig. 3(C), the relationships between the two pairs of j sparse vectors are calculated

$$\mathbf{W} = \exp\left(-\frac{\|\mathbf{C}_{:m} - \mathbf{C}_{:n}\|_2^2}{2\sigma^2}\right) \tag{19}$$

where σ denotes the width parameter of the function, which is set as ten in this article; and m, n are the column numbers of any two columns in the sparse representation matrix $\mathbf{C}_{ij}$.

The spectral clustering operation is then applied using the adjacency matrix $\mathbf{W}$. The Laplacian matrix is first calculated as

$$\mathbf{L} = \mathbf{D}' - \mathbf{W}. \tag{20}$$

In addition, the diagonal matrix $\mathbf{D}'$ is calculated as

$$\mathbf{D}' = \begin{pmatrix} d_1 & \cdots & \cdots \\ \cdots & d_2 & \cdots \\ \vdots & \vdots & \ddots \\ \cdots & \cdots & d_n \end{pmatrix}, \quad d_i = \|\mathbf{C}_{:m} - \mathbf{C}_{:n}\|_2^2. \tag{21}$$

Standardizing the Laplacian matrix as

$$\mathbf{L}' = \mathbf{D}'^{-1/2}\mathbf{L}\mathbf{D}'^{-1/2}. \tag{22}$$

The eigenvectors corresponding to the k_1 smallest eigenvalues are then calculated, and the matrix composed of eigenvectors is normalized by rows to obtain the characteristic matrix $\mathbf{L}^*$. K-means clustering is then utilized, and the clustering result of the HSI is output.

IV. Experiments

In order to show the effectiveness of the proposed multi-objective SSC method for hyperspectral imagery, it was compared with several other methods: the traditional FCM clustering method; an improved differential evolution algorithm-based fuzzy clustering method (FCIDE) [64]; original SSC [17], [18]; affine SSC (A-SSC) [25]; a method we designed that we refer to as Pre-ASSC, which presents the A-SSC after the preparatory FCM clustering; the adaptive multi-objective differential evolution algorithm used for automatic clustering (AFCMDE) [42]; and a regularization parameter estimation method using the minimum distance criterion (RPE_MDC) [67]. The proposed methods are referred to as MOMSSC_L0 and MOMSSC_L0_TV, which means a modified SSC method based on a multi-objective optimization algorithm and a method considering the TV spatial prior. Furthermore, in order to confirm the accuracy of modeling using the l_0 norm, the l_1 norm is used in another version of the multi-objective SSC method, which is referred to as MOMSSC_L1. The tested HSI data were commonly used and publicly available hyperspectral scenes, i.e., a subset from the Indian Pines Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) image (Indian Pines AVIRIS_subset),

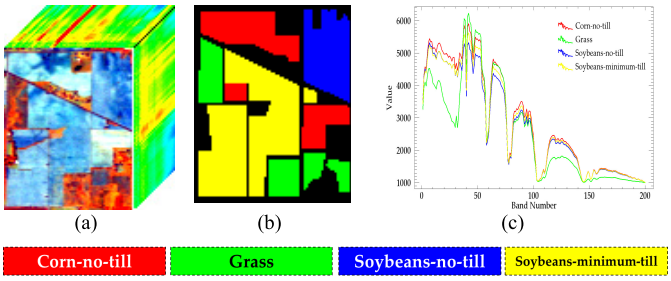


Fig. 11. Indian Pines AVIRIS_subset image. (a) Original image. (b) Ground-truth data. (c) Spectral curves.

a subset from the Salinas AVIRIS image (Salinas AVIRIS_subset), and a subset from the Pavia University Reflective Optics System Imaging Spectrometer (ROSIS) image (PaviaU ROSIS_subset). The value of the fuzzy weighting exponent m was set as 2; the numbers of clusters for SSC were 4, 6, and 7 for the three tested images, respectively; and the value of the width parameter of the function (18) was set as 10. The producer's accuracy (PA), user's accuracy (UA), overall accuracy (OA), and Kappa coefficient are utilized for the quantitative evaluation, and are calculated as follows:

$$\begin{aligned} \mathrm{PA}_j &= \frac{\mathrm{Nu}_{jj}}{\mathrm{Nu}_{+j}}, \quad \mathrm{UA}_i = \frac{\mathrm{Nu}_{ii}}{\mathrm{Nu}_{i+}}, \quad \mathrm{OA} = \frac{\sum_{i=1}^{K} \mathrm{Nu}_{ii}}{\mathrm{Nu}} \\ \mathrm{Kappa} &= \frac{\mathrm{Nu}\sum_{i=1}^{K} \mathrm{Nu}_{ii} - \sum_{i=1}^{K} \mathrm{Nu}_{i+} \times \mathrm{Nu}_{+j}}{\mathrm{Nu}^2 - \sum_{i=1}^{K} \mathrm{Nu}_{i+} \times \mathrm{Nu}_{+j}} \end{aligned} \tag{23}$$

where Nu_{ij} is the number of samples in the test samples that should be the ith class, but in the actual classification results, are designated as the jth class; Nu_{+j} denotes the total number of samples classified as the jth class in the actual classification results; and Nu_{i+} represents the total number of samples of the ith class in the test samples.

A. Experimental Data

1) Indian Pines AVIRIS_subset Image: This HSI scene was gathered by the AVIRIS sensor over the Indian Pines test site in Northwest Indiana. The scene used in this experiment was a subset of this image. After the pretreatment, 20 channels were removed from the original 220 bands, leaving 200 bands. The image subset size was 72 × 91 pixels, as shown in Fig. 11(a). The image subset contained four classes, which numbered 1008, 747, 737, and 1947 pixels, respectively, as shown in Fig. 11(b). The spectral curves are shown in Fig. 11(c). It should be noted that the curves of corn-no-till, soy-beans-minimum-till, and soybeans-no-till are very close, which indicates that they are easily misclassified.

2) Salinas AVIRIS_subset Image: This HSI scene was collected by the 224-band AVIRIS sensor over Salinas Valley, California. The 20 water absorption bands were discarded, leaving 204 bands for the subsequent experiment. A subset of 80 × 100 pixels was cut from the original large image, as shown in Fig. 12(a). Six classes were contained in this Salinas_subset image: broccoli-green-weeds-1/2, soil-vineyard-developed, and lettuce-romaine-4wk/5wk/6wk,

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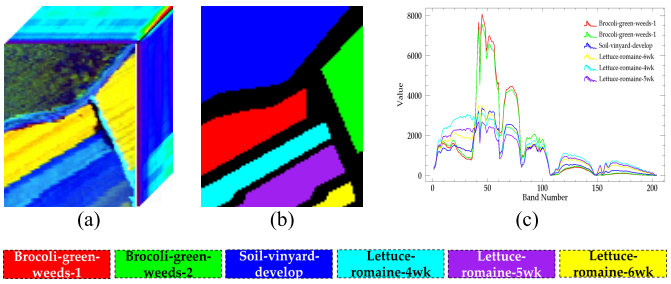


Fig. 12. Salinas AVIRIS_subset image. (a) Original image. (b) Ground-truth data. (c) Spectral curves.

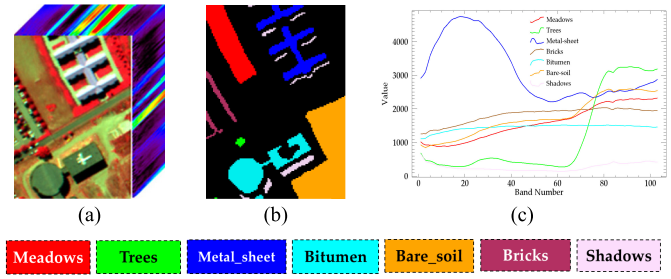


Fig. 13. PaviaU ROSIS_subset image. (a) Original image. (b) Ground-truth data. (c) Spectral curves.

which numbered 989, 714, 2350, 680, 750, and 141 pixels, respectively, as shown in Fig. 12(b).

The spectral curves are shown in Fig. 12(c). It should be noted that the curves of the broccoli-green-weeds-1 and broccoli-green-weeds-2 classes are very close, as are the curves of the soil-vineyard-developed and lettuce-romaine-4wk/5wk/6wk classes, which indicates that they are easily misclassified.

3) PaviaU ROSIS_subset Image: This HSI scene was collected by the 103-band ROSIS sensor over Pavia University, Italy. The size of the original image is 610 × 610 pixels, and a subset sized 100 × 140 pixels was obtained from it for the subsequent experiment, as shown in Fig. 13(a). The subset image contained seven classes: meadows, trees, metal sheet, bare soil, bricks, shadows, and bitumen, which numbered 867, 33, 679, 1759, 188, 255, and 771 pixels, respectively, as shown in Fig. 13(b). From the spectral curves shown in Fig. 13(c), the curves are again very mixed, which means that the pixels are easily misclassified and have a nonlinear distribution.

B. Experiment 1: Indian Pines AVIRIS_subset Image

For this HSI scene, when constructing the new dictionary, the number of classes for the pre-clustering was set as $K = 50$. In addition, the size of the population was set as $P = 30$, and the number of generations was set as 50. The clustering results of the different methods are provided in Fig. 14.

The image in Fig. 14(a) is the result of the pre-clustering, where the new dictionary was constructed by selecting the pixels with the highest degree of membership in each cluster. In addition, for the FCIDE and AFCMDE methods, principal component analysis (PCA) was used to reduce the bands of the Indian Pines AVIRIS_subset image, leaving ten bands for the following experiment. First of all, from the visual point of view, the original SSC method obtains poor results for the HSI clustering, which indicates that the hyperspectral data exhibit a nonlinear distribution in the low-dimensional subspace. The A-SSC, Pre-ASSC, MOMSSC_L1, MOMSSC_L0, and MOMSSC_L0_TV methods perform better on the soybeans-no-till feature class than the methods of FCM and FCIDE. As for the class of soybeans-minimum-till, the four clustering methods—AFCMDE, RPE_MDC, MOMSSC_L0, and MOMSSC_L0_TV—perform better than the others, while AFCMDE, RPE_MDC, and MOMSSC_L0 do not perform, as well as MOMSSC_L0_TV on the corn-no-till and grass classes. Furthermore, it can be found that the proposed method achieves a better balance between the classes than the MOMSSC_L1 method, which indicates the advantage of modeling using the l_0 norm. In addition, the introduction of the spatial prior term makes the clustering result smoother and less noisy. Overall, MOMSSC_L0_TV achieves the best visual effect among all the methods.

In order to further verify the effectiveness of the proposed method, the quantitative evaluation of the different clustering results is presented in Table I, where the highest accuracy in each row is marked in bold. It can be clearly seen that SSC performs the worst on the classes of corn-no-till (43.15%) and soy-beans-minimum-till (51.72%), which is consistent with the visual results. In addition, FCM performs the worst on the class of soybeans-no-till (11.67%), and FCIDE performs the worst on the class of grass (60.37%). The proposed bi-objective optimization-based method—MOMSSC_L0—performs the best on the class of soybeans-no-till (69.74%), which is easily confused due to the spectral curves being very close. In addition, the proposed three-objective optimization-based method—MOMSSC_L0_TV—performs the best on the class of grass (97.99%). This indicates that MOMSSC_L0 and MOMSSC_L0_TV can better mine the spectral-spatial information in the HSI than the other methods. For the evaluation of the OA, the proposed method—MOMSSC_L0_TV—obtains the highest value, i.e., 73.73%, which is 12.66%, 9.62%, 15.27%, 10.54%, 8.96%, 5.22%, 3.6%, 2.97%, and 1.91% more than the other nine methods, respectively. Furthermore, the result of the method—RPE_MDC—is slightly worse than that of MOMSSC_L1 and MOMSSC_L0, which means that the methods which do not require the determination of the regularization parameter are superior to the method that does require the determination of the regularization parameter. Overall, the proposed method—MOMSSC_L0_TV—achieves the best clustering performance with the Indian Pines AVIRIS_subset image.

C. Experiment 2: Salinas AVIRIS_subset Image

For this HSI scene, when constructing the new dictionary, the number of classes for the pre-clustering was set as $K = 30$. In addition, the size of the population was set as $P = 30$, and the number of generations was set as 50. The clustering results of the different methods are provided in Fig. 15 for the Salinas AVIRIS_subset image.

For the FCIDE and AFCMDE methods, the pretreatment of the Salinas AVIRIS_subset image was the same as for

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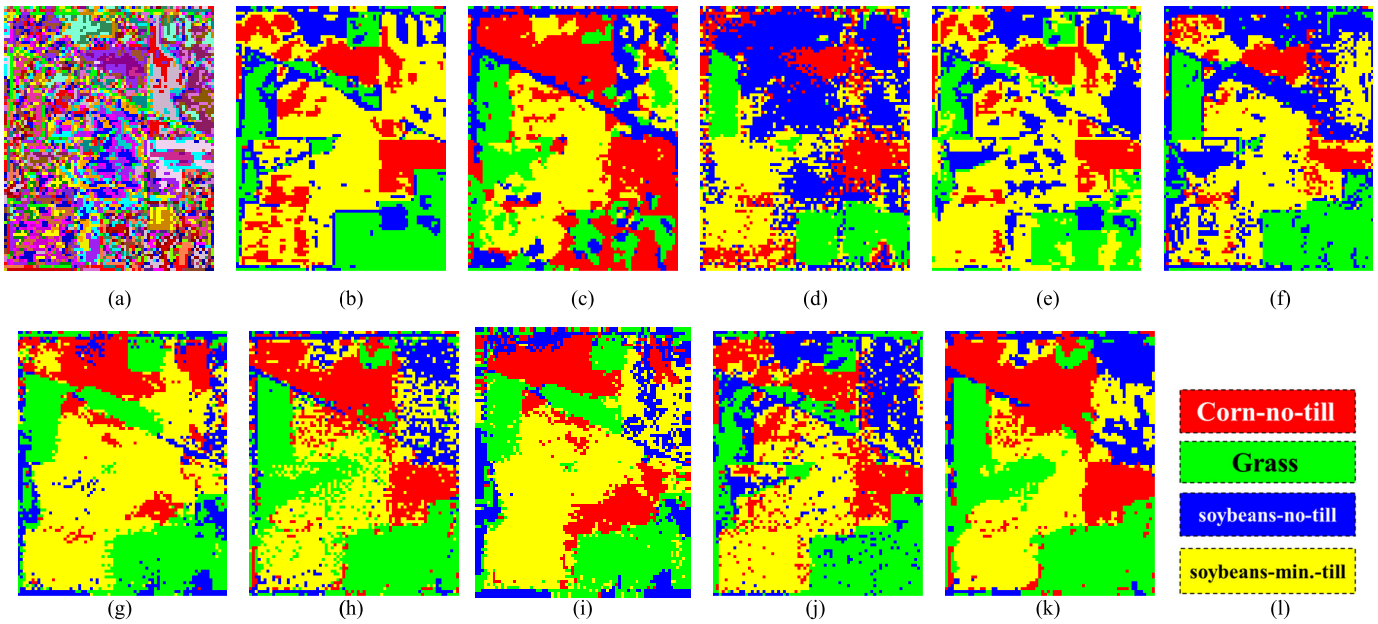


Fig. 14. Clustering results of the different methods for the Indian Pines AVIRIS_subset image. (a) Result of the preclustering. (b) FCM. (c) FCIDE. (d) SSC. (e) A-SSC. (f) Pre-ASSC. (g) AFCMDE. (h) MOMSSC_L1. (i) RPE_MDC. (j) MOMSSC_L0. (k) MOMSSC_L0_TV. (l) Legend.

TABLE I

COMPARISON OF THE CLUSTERING ACCURACIES FOR THE INDIAN PINES AVIRIS_SUBSET IMAGE

	Class	1	2	3	4	5	6	7	8	9	10
PA (%)	Corn-no-till	61.11	**79.76**	43.15	52.48	44.05	52.08	73.51	74.01	64.29	79.17
	Grass	73.09	60.37	93.17	67.20	95.45	94.38	97.46	91.43	77.38	**97.99**
	Soybeans-no-till	11.67	38.81	62.01	56.04	49.39	20.02	53.46	25.10	**69.74**	53.87
	Soybeans-minimum-till	75.14	67.03	51.72	69.90	69.54	**85.41**	64.20	80.48	74.37	69.13
UA (%)	Corn-no-till	57.73	65.79	54.31	68.43	74.00	59.59	78.58	63.92	63.78	**76.66**
	Grass	97.85	48.60	**99.86**	97.67	98.89	85.45	58.62	86.46	94.14	65.01
	Soybeans-no-till	15.61	50.98	26.87	34.79	35.72	55.85	78.64	**89.81**	51.50	85.93
	Soybeans-minimum-till	64.65	75.52	**81.21**	69.26	64.51	67.38	71.31	67.66	79.96	74.36
	OA (%)	61.07	64.11	58.46	63.19	64.77	68.51	70.13	70.76	71.82	**73.73**
	Kappa	0.4276	0.4950	0.4435	0.4760	0.4941	0.5299	0.5809	0.5870	0.6022	**0.6290**

Note: 1-FCM. 2-FCIDE. 3-SSC. 4-A-SSC. 5-Pre-ASSC. 6-AFCMDE. 7-MOMSSC_L1. 8-RPE_MDC. 9-MOMSSC_L0. 10-MOMSSC_L0_TV.

the Indian Pines AVIRIS_subset image, leaving ten bands. From the visual point of view, the result of the original SSC method is clearly worse than that of FCM, FCIDE, and A-SSC, especially in the class of broccoli-green-weeds-2, where many pixels are misclassified as broccoli-green-weeds-1. The Pre-ASSC, AFCMDE, MOMSSC_L1, RPE_MDC, MOMSSC_L0, and MOMSSC_L0_TV methods perform better than the four other methods on the broccoli-green-weeds-2 class. However, among the results of these five methods, the performances with the broccoli-green-weeds-1/2, soil-vineyard-developed, and lettuce-romaine-4wk/5wk classes are very similar, while MOMSSC_L0 and MOMSSC_L0_TV perform better than Pre-ASSC, AFCMDE, MOMSSC_L1, and RPE_MDC on the lettuce-romaine-6wk class. Furthermore, the introduction of the spatial prior term in MOMSSC_L0_TV makes the clustering result smoother. Overall, the proposed method—MOMSSC_L0_TV—performs the best of all the methods at the visual level.

In order to further verify the effectiveness of the proposed method, a quantitative evaluation is provided in Table II, where the evaluation indicators are the same as for the Indian Pines AVIRIS_subset image. First of all, the original SSC method performs the worst in the two classes of broccoli-green-weeds-2 (only 15.27%) and lettuce-romaine-6wk (only 13.48%). This performance of the SSC method confirms once again that the HSI data present a nonlinear distribution in the low-dimensional subspace. From the PA, it can be found that RPE_MDC obtains the best performance on the classes of broccoli-green-weeds-2 and lettuce-romaine-4wk, but it obtains a very poor performance on the lettuce-romaine-5wk class. The methods of MOMSSC_L0 and MOMSSC_L0_TV achieve the best performances on the classes of soil-vineyard-developed and lettuce-romaine-6wk. For the OA, MOMSSC_L0_TV achieves the highest value of 96.75%, which is 5.94%, 5.87%, 11.47%, 5.14%, 5.32%, 5.98%, 4.50%, 3.67%, and 1.32% more than the other nine methods, respectively. Overall, the proposed method—MOMSSC_L0_TV—achieves the best clustering result with the Salinas AVIRIS_subset image, both visually and quantitatively.

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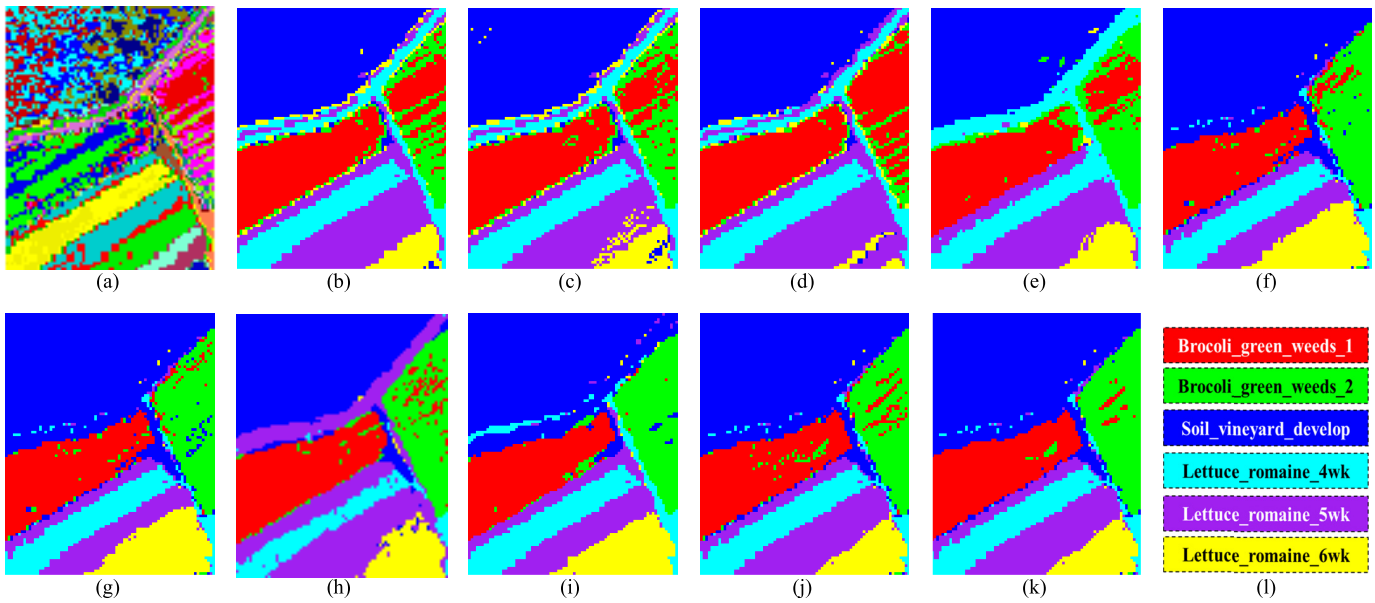


Fig. 15. Clustering results of the different methods for the Salinas AVIRIS_subset image. (a) Result of the pre-clustering. (b) FCM. (c) FCIDE. (d) SSC. (e) A-SSC. (f) Pre-ASSC. (g) AFCMDE. (h) MOMSSC_L1. (i) RPE_MDC. (j) MOMSSC_L0. (k) MOMSSC_L0_TV. (l) Legend.

TABLE II

COMPARISON OF THE CLUSTERING ACCURACIES FOR THE SALINAS AVIRIS_SUBSET IMAGE

	Class	1	2	3	4	5	6	7	8	9	10
PA (%)	Brocoli-green-weeds-1	98.99	97.88	**99.60**	97.78	95.96	95.96	**99.60**	97.20	94.84	97.78
	Brocoli-green-weeds-2	44.96	57.28	15.27	60.50	91.32	92.16	89.92	**95.85**	87.68	93.98
	Soil-vinyard-develop	97.15	97.02	97.83	97.49	**99.87**	99.68	98.98	99.06	**99.87**	**99.87**
	Lettuce-romaine-4wk	93.24	93.24	93.24	86.03	92.35	92.42	68.24	**94.56**	92.35	92.35
	Lettuce-romaine-5wk	**100.00**	93.20	**100.00**	98.13	56.67	50.93	84.00	64.13	91.60	91.60
	Lettuce-romaine-6wk	99.29	85.82	13.48	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
UA (%)	Brocoli-green-weeds-1	71.36	76.04	61.95	77.42	94.52	**97.63**	93.19	96.12	91.42	95.74
	Brocoli-green-weeds-2	96.98	95.12	96.46	92.11	97.02	97.05	99.38	**99.89**	92.74	97.11
	Soil-vinyard-develop	99.96	99.13	99.61	**100.00**	98.74	97.67	99.27	98.64	99.79	99.79
	Lettuce-romaine-4wk	98.45	98.45	98.45	93.75	99.84	99.84	**100.00**	97.57	99.85	99.84
	Lettuce-romaine-5wk	92.71	92.22	79.87	88.57	89.29	88.21	72.83	91.97	**93.09**	**93.09**
	Lettuce-romaine-6wk	76.09	55.25	63.33	**88.13**	30.26	27.70	56.63	34.31	69.12	69.12
	OA (%)	90.81	90.88	85.28	91.61	91.43	90.77	92.25	93.08	95.43	**96.75**
	Kappa	0.8768	0.8779	0.8010	0.8875	0.8857	0.8769	0.8962	0.9078	0.9387	**0.9564**

Note: 1-FCM. 2-FCIDE. 3-SSC. 4-A-SSC. 5-Pre-ASSC. 6-AFCMDE. 7-MOMSSC_L1. 8-RPE_MDC. 9-MOMSSC_L0. 10-MOMSSC_L0_TV.

D. Experiment 3: PaviaU ROSIS_subset Image

For this HSI scene, the parameters were set as $K = 50$ and $P = 30$, and the number of generations was set as 50. The clustering results of the different methods are provided in Fig. 16.

For the AFCMDE method, the pretreatment of the PaviaU ROSIS_subset image was the same as for the former two images, leaving ten bands. From the visual point of view, the A-SSC, Pre-ASSC, AFCMDE, MOMSSC_L1, RPE_MDC, MOMSSC_L0, and MOMSSC_L0_TV methods perform better than the other three methods on the class of metal sheet. For the meadows and bare soil classes, there are many misclassified pixels in the results of all the methods. Among the comparison methods, FCM, FCIDE, RPE_MDC, MOMSSC_L0, and MOMSSC_L0_TV perform better than the other methods on the bare soil class. In addition, it should be noted that the pixels of the bricks class are misclassified into the bitumen class by all the methods. As for the bitumen and shadows classes, they show good performances with all the methods, because there is no intersection in their spectral curves. In addition, one phenomenon should be noted, in that there are fewer outliers in the result of AFCMDE than in the result of the proposed method. A possible reason for this is that, in AFCMDE, its two objective functions are the Jm function used in FCM and the Xie-Beni (XB) index, and thus it is similar to FCM, but obtains more misclassifications than the proposed method. In addition, it can be found that the proposed method achieves a better balance between the classes than the MOMSSC_L1 method, which indicates the advantage of modeling using the l_0 norm. Furthermore, due to the introduction of the spatial prior term, the proposed method—MOMSSC_L0_TV—obtains the best visual performance among all the comparison methods.

In order to further verify the effectiveness of the proposed method, a quantitative evaluation for the PaviaU ROSIS_subset image is provided in Table III. For the original SSC method,

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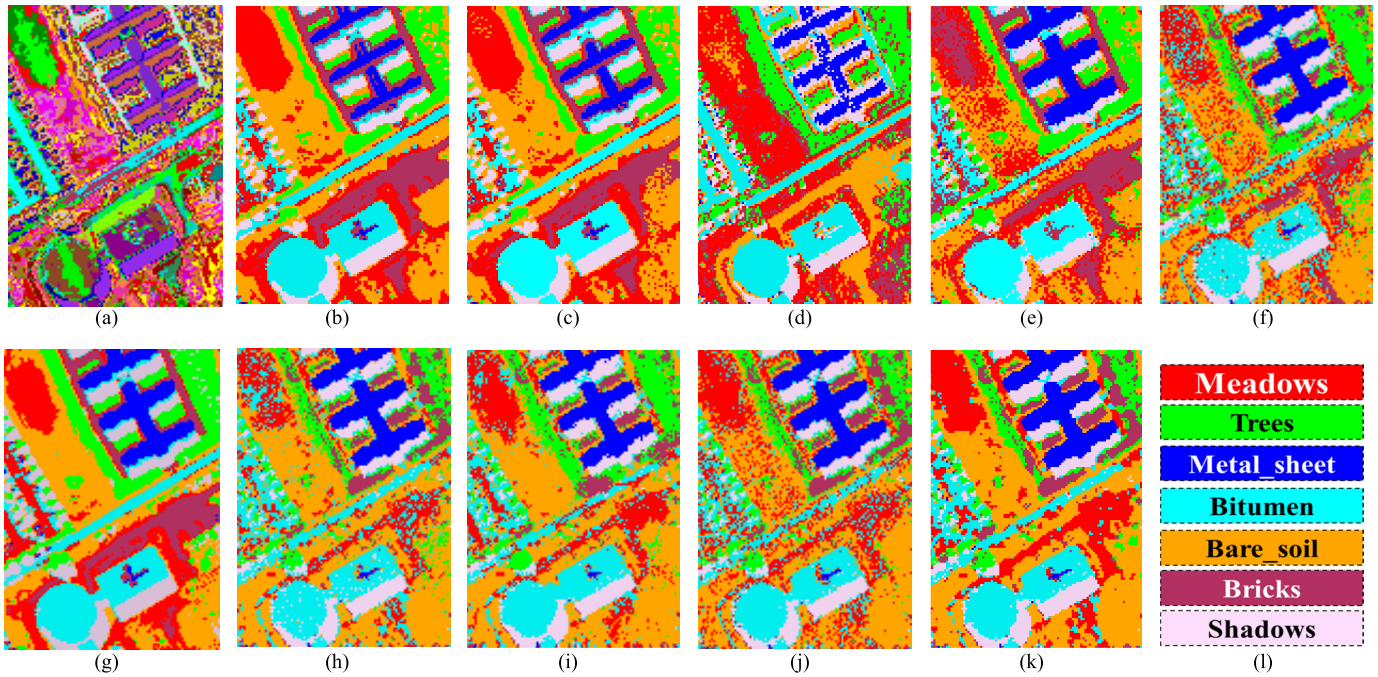


Fig. 16. Clustering results of the different methods for the PaviaU ROSIS_subset image. (a) Result of the pre-clustering. (b) FCM. (c) FCIDE. (d) SSC. (e) A-SSC. (f) Pre-ASSC. (g) AFCMDE. (h) MOMSSC_L1. (i) RPE_MDC. (j) MOMSSC_L0. (k) MOMSSC_L0_TV. (l) Legend.

TABLE III
COMPARISON OF THE CLUSTERING ACCURACIES FOR THE PAVIAU ROSIS_SUBSET IMAGE

Classes		1	2	3	4	5	6	7	8	9	10
PA (%)	Meadows	72.20	71.97	**80.39**	43.94	55.36	62.28	45.79	49.25	63.67	68.17
	Trees	66.67	66.67	30.30	**100.00**	75.76	87.88	75.76	**100.00**	75.76	78.79
	Metal sheet	63.92	70.54	67.01	99.41	99.71	97.20	99.71	99.71	99.71	**100.00**
	Bitumen	**99.87**	**99.87**	98.70	99.22	89.49	99.74	90.66	91.96	95.46	95.85
	Bare soil	43.49	47.36	35.65	45.71	54.75	44.68	59.86	**62.19**	55.71	60.49
	Bricks	0.53	0.53	0.00	**39.89**	0.00	0.00	0.00	0.00	0.00	0.00
	Shadows	**100.00**	**100.00**	78.43	95.29	99.61	**100.00**	99.61	**100.00**	99.61	**100.00**
UA (%)	Meadows	51.57	50.90	**64.30**	52.48	46.38	50.42	48.65	52.07	49.60	48.24
	Trees	75.86	**81.48**	4.63	30.56	18.94	22.14	18.94	26.40	19.69	32.50
	Metal sheet	**100.00**	**100.00**	99.56	**100.00**	**100.00**	99.70	**100.00**	**100.00**	**100.00**	**100.00**
	Bitumen	87.90	87.90	75.72	86.73	61.12	**97.34**	58.54	59.68	68.47	82.11
	Bare soil	75.15	76.56	73.94	**85.81**	79.32	70.56	76.97	78.54	79.74	79.94
	Bricks	0.14	0.17	0.00	**7.68**	0.00	0.00	0.00	0.00	0.00	0.00
	Shadows	**100.00**	**100.00**	46.84	98.38	79.13	79.94	79.13	81.99	89.12	88.54
OA (%)		63.12	65.55	60.41	65.38	67.86	66.76	68.21	70.19	70.83	**73.68**
Kappa		0.5425	0.5691	0.5172	0.5794	0.5905	0.5851	0.5917	0.6167	0.6273	**0.6608**

Note: 1-FCM. 2-FCIDE. 3-SSC. 4-A-SSC. 5-Pre-ASSC. 6-AFCMDE. 7-MOMSSC_L1. 8-RPE_MDC. 9-MOMSSC_L0. 10-MOMSSC_L0_TV.

the PAs of the trees, bare soil, bricks, and shadows classes are the lowest among all the methods. In addition, the RPE_MDC method obtains the best performance on the classes of trees, bare soil, and shadows, but it obtains a very poor result for the meadows class. Furthermore, the proposed method—MOMSSC_L0_TV—obtains the maximum number of correct classifications for the metal sheet and shadows classes (100.00%), which is consistent with the visual effect. All the clustering methods obtain nearly 100% correct classifications for the bitumen and shadows classes, and almost 0% correct classifications for the bricks class, which is consistent with the visual effect. For the OA, the proposed method—MOMSSC_L0_TV—obtains the highest value of 73.68%, which is 10.56%, 8.13%, 13.27%, 8.30%, 5.82%, 6.92%, 5.47%, 3.49%, and 2.85% more than the other nine methods, respectively. Overall, MOMSSC_L0_TV achieves the best clustering result, both visually and quantitatively.

E. Experiment 4: Test With a Larger Image: The Pavia ROSIS Image

In order to further verify the effectiveness of the proposed method, a larger hyperspectral remote sensing image was applied. The HSI was a scene acquired by the ROSIS sensor during a flight campaign over Pavia, northern Italy. The image has 102 bands, and the full image size of 450 × 450 pixels was used in this experiment. Fig. 17(a) and (b) shows the

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original image and the corresponding ground-truth image, in which seven classes are contained. The seven methods used in this experiment were FCM, FCIDE, SSC, MOMSSC_L1, RPE_MDC, MOMSSC_L0, and MOMSSC_L0_TV.

In this scene, the number of classes for the pre-clustering was set as $K = 60$. The size of the population was set as $P = 30$, and the number of generations was set as 50. The results for the Pavia ROSIS image are shown in Fig. 17(d)–(j). Firstly, from the visual point of view, the FCM method obtains the most outliers in the class of water. In addition, for the black window labeled as 1, which should be the bare soil class, in the methods of FCM, FCIDE, and SSC, it has been misclassified as the bricks and building classes. As for the black window labeled as 2, which should be the building class, in the methods of FCM and MOMSSC_L1, it has been misclassified as the bricks class and the asphalt class, respectively. Lastly, for the black window labeled as 3, it has been misclassified as the building and bare soil classes in the SSC, MOMSSC_L1, and RPE_MDC methods, whereas it should be the bricks class. Overall, the proposed method obtains a better visual result than the other methods.

Similarly, for the quantitative evaluation, the PA, UA, OA, and Kappa coefficient values are listed in Table IV. It can be found that MOMSSC_L0_TV obtains the best OA of 93.72%, which represents gains of 3.08%, 2.97%, 4.77%, 1.54%, 1.24%, and 0.63% over FCM, FCIDE, SSC, MOMSSC_L1, RPE_MDC, and MOMSSC_L0, respectively. Furthermore, it achieves the maximum number of correct classifications in the classes of asphalt and bricks, which is consistent with the visual result. Overall, MOMSSC_L0_TV can obtain a better balance between the classes in this large HSI, which confirms the superiority of the multi-objective sparse optimization strategy, modeling using the l_0 norm, and the spatial term.

V. Parameter Analyses

A. Sensitivity Analysis for the Number of Classes in the Pre-Clustering Process

The number of classes for the pre-clustering via FCM should be determined in advance. This parameter is important due to its size directly determining the form of the data dictionary in the sparse optimization. For the four tested hyperspectral remote sensing images, the class number was set as 50, 30, 50, and 60, respectively. In fact, the numbers of classes in the four experiments were set by visually pre-analyzing the distributions and the complexities of the four hyperspectral remote sensing images. To the best of our knowledge, the more mixed the distribution and the more complex the structure, the higher the class number should be set for the pre-clustering. For a more specific understanding of the effect of this parameter, a sensitivity analysis is provided in the following. Class numbers of 20, 30, 40, 50, and 60 were used to test the effect on the OA of the final clustering results for the four hyperspectral remote sensing images, as shown in Fig. 18. Clearly, it can be seen that the change in the number of classes in the pre-clustering process results in some fluctuations in the final clustering accuracy, but the range of these fluctuations is small. Thus, the pre-clustering operation and the novel strategy for constructing the dictionary can be considered to be effective.

In Fig. 18(a), for the Indian Pines AVIRIS_subset image, the highest value of OA is obtained when $K = 40$, and the overall trend first increases and then decreases. For the Salinas AVIRIS_subset image in Fig. 18(b), the change is slightly more complex than for the Indian Pines AVIRIS_subset image, and the highest value of OA is obtained when $K = 40$. For the Salinas AVIRIS_subset image in Fig. 18(c), this shows an overall upward trend, and the highest value of OA is obtained when $K = 60$. Lastly, for the Pavia ROSIS image in Fig. 18(d), this shows an overall upward trend, and the highest value of OA is obtained when $K = 60$. Thus, for the proposed method of MOMSSC_L0, the final results were obtained using these values.

B. Sensitivity Analysis for the Population Size P

The population size is an important parameter in the evolutionary multi-objective sparse optimization. The number of individuals in the population affects the evolution of the entire population. Thus, further experiments were undertaken to test the effect on the clustering accuracy of the size of the population. The test results are shown in Fig. 19, where different values of population size P were tested from 10 to 60.

Generally speaking, when the size of the population is too small, the clustering results deteriorate. For the Indian Pines AVIRIS_subset image and the PaviaU ROSIS_subset image, the performances are similar: when the size of the population is 40, the OAs are the highest; however, the OAs of the clustering results show a slight decrease as the number of individuals increases, due to the not very effective evolution in the population. In addition, for the Salinas AVIRIS_subset and Pavia ROSIS images, the values of the OA continue to increase with the increase of the number of individuals in the population but the increased speed tends to be gentle in the later stages. All in all, from the numerical point of view, the OA is only slightly influenced by the size of the population, which indicates that the proposed approach shows good robustness.

C. Analysis of the Influence of the Initialization Strategy

In order to confirm the effectiveness and generality of the random initialization for the individuals in a population, a comparison of different initialization strategies was conducted. Three different initialization strategies were considered: random initialization, uniform initialization, and initialization with the same value. The proposed bi-objective optimization-based method—MOMSSC_L0—was used for the testing with the Indian Pines AVIRIS_subset, Salinas AVIRIS_subset, and PaviaU ROSIS_subset images. The results are provided in Fig. 20. It can be found that the final clustering OAs of the three different initialization strategies are very close for the three tested images, which indicates that the strategy of random initialization used in the proposed method is effective.

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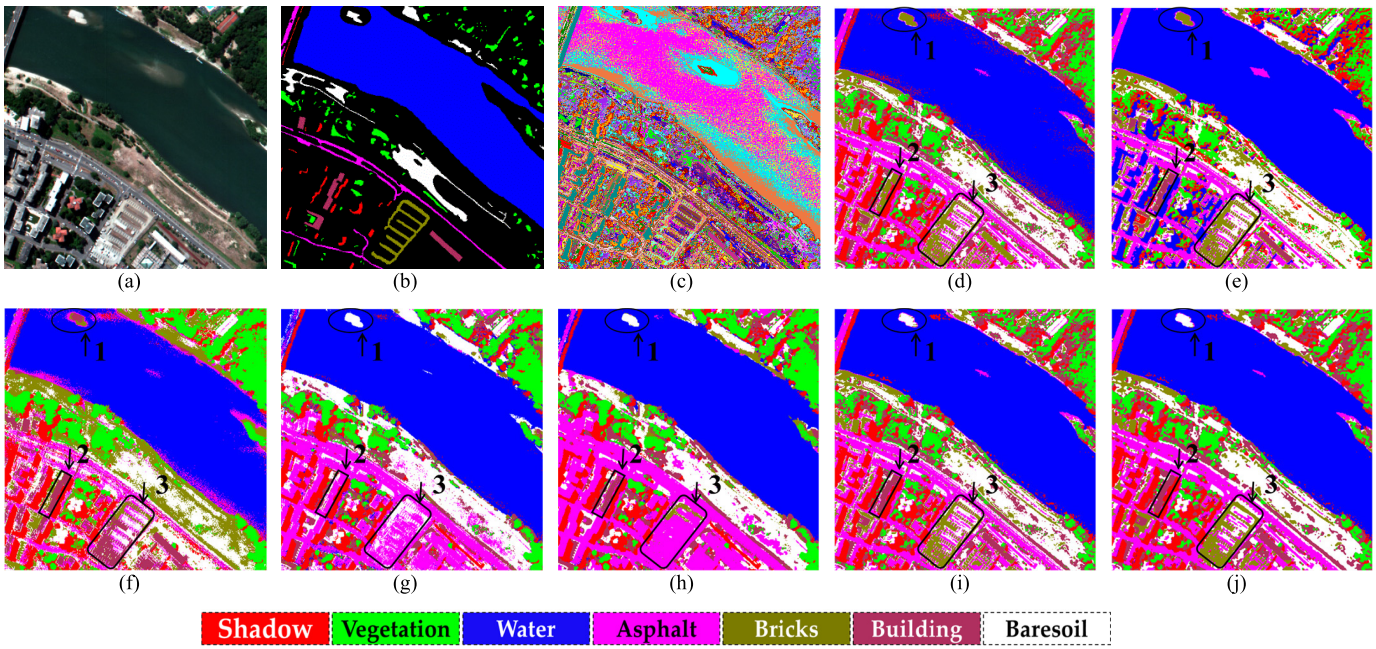


Fig. 17. Clustering results of the different methods for the Pavia ROSIS image. (a) Original image. (b) Ground-truth image. (c) Result of the pre-clustering. (d) FCM. (e) FCIDE. (f) SSC. (g) MOMSSC_L1. (h) RPE_MDC. (i) MOMSSC_L0. (j) MOMSSC_L0_TV.

TABLE IV
COMPARISON OF THE CLUSTERING ACCURACIES FOR THE PAVIA ROSIS IMAGE

Class	PA (%)							UA (%)						
	1	2	3	4	5	6	7	1	2	3	4	5	6	7
Shadow	99.11	0.07	96.12	99.63	**99.85**	99.11		99.11	0.09	64.29	93.43	**99.63**	86.39	90.53
Vegetation	73.51	71.96	**77.51**	63.15	61.01	73.61	74.51	73.51	99.77	99.97	99.89	99.88	**100.00**	**100.00**
Water	98.49	99.44	98.23	99.82	**99.95**	99.45	99.63	99.98	97.85	99.96	99.41	99.88	**100.00**	99.98
Asphalt	99.18	98.69	70.01	90.83	98.10	99.20	**99.74**	85.48	83.96	64.09	51.32	52.35	85.45	**88.64**
Bricks	74.21	79.91	28.27	0.00	23.32	74.21	**84.21**	25.56	30.08	11.46	0.00	29.49	32.76	**36.12**
Building	0.05	77.58	**84.71**	0.00	38.90	61.80	62.67	0.08	**99.51**	52.53	0.00	33.50	48.31	50.54
Bare soil	52.49	42.68	38.77	**94.94**	76.08	56.20	58.08	52.49	89.71	77.55	82.86	95.49	92.53	**96.09**
OA (%)	90.64	90.75	88.95	92.18	92.48	93.09	**93.72**							
Kappa	0.7789	0.7724	0.7402	0.8093	0.8176	0.8348	**0.8495**							

Note: 1-FCM. 2-FCIDE. 3-SSC. 4-MOMSSC_L1. 5-RPE_MDC. 6-MOMSSC_L0. 7-MOMSSC_L0_TV.

D. Analysis of the Convergence Condition

In order to perform an analysis of the convergence condition, visualization of the convergence condition for the final solution set is presented in this part. Illustrations of the convergence conditions for the Indian Pines AVIRIS_subset image and Salinas AVIRIS_subset image are presented in Fig. 21(a) and (b). From the convergence results, it can be found that the individuals in a population are nondominated in the final solution set, and the final solution is selected for the subsequent processing through the knee point-based method. In addition, in the proposed method, the problem we are facing is the clustering of hyperspectral remote sensing imagery, and the Pareto optimal set is not guaranteed for the results of the multi-objective optimization. However, from the comparison of the experimental methods, we believe that the multi-objective optimization is sufficient to deal with the application of hyperspectral remote sensing image processing.

E. Analysis of the Computation Time

A computation time comparison for the eight clustering methods (FCM, FCIDE, SSC, A-SSC, Pre-ASSC, AFCMDE, MOMSSC_L1, and MOMSSC_L0) with the three hyperspectral remote sensing images (Indian Pines AVIRIS_subset, Salinas AVIRIS_subset, and PaviaU ROSIS_subset images) is presented in Table V. Firstly, the traditional fuzzy clustering approach, i.e., FCM, requires the least computation time, but the clustering accuracy is not satisfactory. Meanwhile, the FCIDE method requires much more computation time, while obtaining a better result than FCM. Then, for the SSC, A-SSC, and Pre-ASSC methods, it can be seen that the computation time has been greatly reduced through the use of the smaller data dictionary and the sparse representation matrix. However, the evolutionary multi-objective optimization-based methods (AFCMDE, MOMSSC_L1, and MOMSSC_L0) require more computation time for clustering the remote sensing images, while MOMSSC_L0 obtains the

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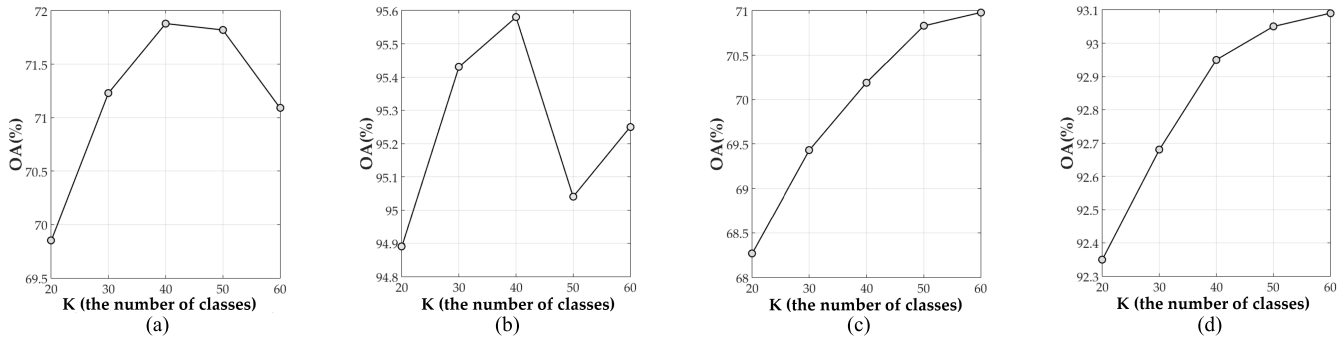


Fig. 18. Effect of K in the pre-clustering. (a) Indian Pines AVIRIS_subset image. (b) Salinas AVIRIS_subset image. (c) PaviaU ROSIS_subset image. (d) Pavia ROSIS image.

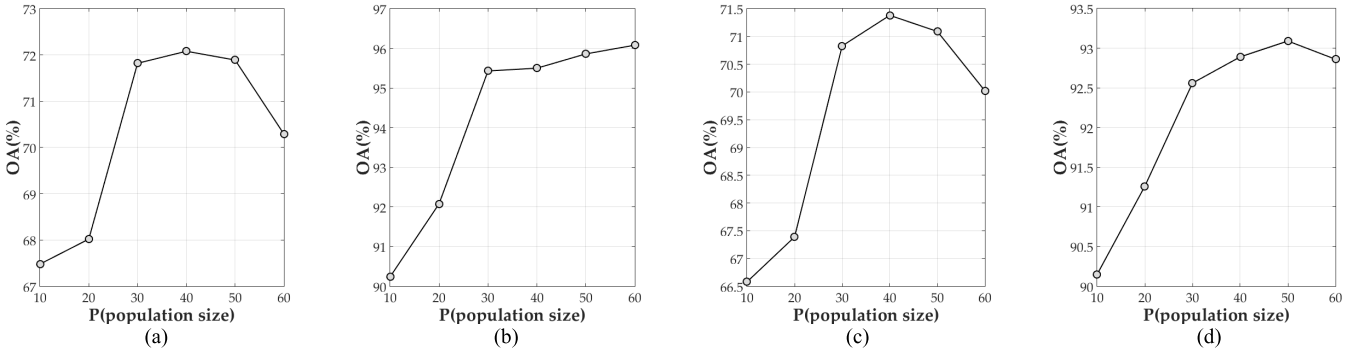


Fig. 19. Effect on the OA of the population size. (a) Indian Pines AVIRIS_subset image. (b) Salinas AVIRIS_subset image. (c) PaviaU ROSIS_subset image. (d) Pavia ROSIS image.

TABLE V
COMPUTATION TIME COMPARISON FOR THE DIFFERENT METHODS WITH THE THREE HSIS

Methods		1	2	3	4	5	6	7	8	9	10
Time (s)	A	5.71	3.578E3	4.526E3	1.612E3	4.743E3	2.240E2	6.046E3	2.043E4	2.048E4	2.660E3
	B	7.35	5.783E3	6.230E3	2.206E3	6.345E3	3.370E2	7.024E3	2.432E4	2.435E4	3.204E3
	C	11.14	9.846E3	1.043E4	3.898E3	1.090E4	1.502E3	1.125E4	2.706E4	2.708E4	3.395E3

Note: 1-FCM. 2-FCIDE. 3-SSC. 4-SSC_P. 5-A-SSC. 6-Pre-ASSC. 7-AFCMDE. 8-MOMSSC_L1. 9-MOMSSC_L0. 10-MOMSSC_L0_P. A-Indian Pines AVIRIS_subset image. B-Salinas AVIRIS_subset image. C-PaviaU ROSIS_subset image.

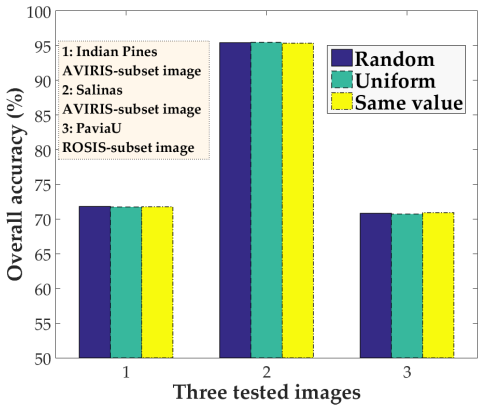


Fig. 20. Analysis of the influence of the different initialization strategies.

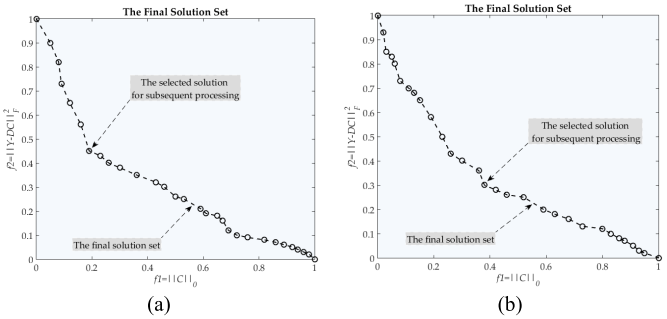


Fig. 21. Analysis of the convergence condition. (a) Indian Pines AVIRIS_subset image. (b) Salinas AVIRIS_subset image.

best clustering performance. Furthermore, considering the good parallelism of the EAs [66], we could implement parallel computing for each individual in a population or conduct multi-objective sparse optimization for multiple pixels at the same time. In this article, we used a personal computer equipped with eight central processing units (CPUs) (Intel(R) Xeon(R) CPU E3-1240 v6 @ 3.70 GHZ), and thus CPU parallel computing could be applied so as to compare the computation time (10-MOMSSC_LO_P). In addition, due to the fact that the ADMM used in SSC can also consider parallel computing, but the decomposition of the subproblems can only be calculated separately, the SSC method proposed in this article can only be decomposed into three subproblems for

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parallel computing (4-SSC_P). From the results in Table V, it can be found that the computational costs for the three tested images can be greatly reduced by the use of CPU parallel computing. Overall, with the rapid development of hardware devices and the fact that EAs have good parallelism, parallel computing could be applied to this clustering algorithm for hyperspectral remote sensing imagery. Therefore, we believe that the proposed method is very extensible.

VI. Conclusion

In this article, two multi-objective SSC methods (bi-objective optimization: MOMSSC_L0; three-objective optimization: MOMSSC_L0_TV) have been proposed for the clustering of hyperspectral remote sensing imagery. In the proposed methods, in order to obtain a better sparse representation matrix, multi-objective optimization theory is introduced to resolve the regularization parameter determination problem. In addition, we directly optimize the l_0 norm as the sparse term, which allows more accurate modeling and reduces the error caused by the convex relaxation of the other norms, such as the l_1 norm. The bi-objective optimization approach is utilized to simultaneously optimize the sparse term and the data fidelity term of the SSC model. The TV spatial prior term is introduced as the third objective function in the three-objective optimization method, MOMSSC_L0_TV. In addition, dictionary preprocessing is used in the proposed algorithm to improve the optimization capability and efficiency, and a knee point-based selection strategy is used to select the final optimal sparse optimization solution from the Pareto optimal solution set. The experimental results obtained in this article confirmed that MOMSSC_L0 and MOMSSC_L0_TV achieve superior clustering performances, and MOMSSC_L0_TV outperforms MOMSSC_L0.

HSI clustering has always been a key task in remote sensing image processing, and how to fully mine the information in HSIs has become one of the main research focuses. To date, the clustering accuracy for HSIs is still low, and the size of the images used for the testing is also relatively small. Thus, how to fully mine the image information and how to cluster larger HSIs will be addressed in our future work. In addition, the different types of HSIs also need to be tested, and the relevant applications will also be considered in our future work; for example, the use of unmanned aerial vehicle (UAV)-borne hyperspectral images [68] for disaster monitoring [69].

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Yuting Wan received the B.S. degree in surveying and mapping engineering from the School of Geosciences and Info-Physics, Central South University, Changsha, China, in 2017. He is currently pursuing the Ph.D. degree in photogrammetry and remote sensing with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan, China.

His research interests include remote sensing image processing, clustering and sparse representation, evolutionary computation, and evolutionary multiobjective optimization.



Yanfei Zhong (M'11–SM'15) received the B.S. degree in information engineering and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2002 and 2007, respectively.

Since 2010, he has been a Full Professor with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University. He organized the Intelligent Data Extraction, Analysis and Applications of Remote Sensing (RSIDEA) Research Group. He has published more than 100 research articles in international journals, including *Remote Sensing of Environment*, *ISPRS Journal of Photogrammetry and Remote Sensing*, and the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. His research interests include hyperspectral remote sensing information processing, high-resolution remote sensing image understanding, and geoscience interpretation for multisource remote sensing data and applications.

Dr. Zhong is a fellow of the Institution of Engineering and Technology (IET). He was a recipient of the 2016 Best Paper Theoretical Innovation Award from the International Society for Optics and Photonics (SPIE). He won the Second-Place Prize in 2013 IEEE GRSS Data Fusion Contest and the Single-view Semantic 3-D Challenge of the 2019 IEEE GRSS Data Fusion Contest, respectively. He is currently serving as an Associate Editor for the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, and the *International Journal of Remote Sensing*.



Ailong Ma received the B.S. degree in surveying and mapping engineering from the China University of Petroleum, Qingdao, China, in 2010, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2017.

He is currently a Research Associate with Wuhan University. His major research interests are remote sensing image processing, evolutionary computing, and deep learning.



Liangpei Zhang (M'06–SM'08–F'19) received the B.S. degree in physics from Hunan Normal University, Changsha, China, in 1982, the M.S. degree in optics from the Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, China, in 1988, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 1998.

From 2011 to 2016, he was a Principal Scientist with the China State Key Basic Research Project appointed by the Ministry of National Science and Technology of China to lead the Remote Sensing Program, China. He is currently a Chang-Jiang Scholar Chair Professor appointed by the Ministry of Education of China in state Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing (LIESMARS), Wuhan University. He has published more than 700 research articles and five books. He is the Institute for Scientific Information (ISI) highly cited author. He holds over 15 patents. His research interests include hyperspectral remote sensing, high-resolution remote sensing, image processing, and artificial intelligence.

Dr. Zhang is a fellow of the Institute of Electrical and Electronic Engineers (IEEE) and the Institution of Engineering and Technology (IET). He was a recipient of the 2010 Best Paper Boeing award, the 2013 Best Paper ERDAS award from the American Society of Photogrammetry and Remote Sensing (ASPRS), and the 2016 Best Paper Theoretical Innovation Award from the International Society for Optics and Photonics (SPIE). His research teams won the top three prizes of the IEEE GRSS 2014 Data Fusion Contest, and his students have been selected as the winners or finalists of the IEEE International Geoscience and Remote Sensing Symposium (IGARSS) student paper contest in recent years. He is the Founding Chair of the IEEE Geoscience and Remote Sensing Society (GRSS) Wuhan Chapter. He is currently serving as an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. He also serves as an Associate Editor or Editor for more than ten international journals.

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